MOLECULAR GENETICS

Module map

- Module 07: Molecular Genetics
- Connected lab: lab-07_molecular-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 07

Molecular Genetics

1. DNA structure
2. Replication
3. Transcription
4. Translation
5. Mutation and gene expression evidence

Lab evidence

lab-07_molecular-genetics.md

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Learning objectives

- Describe DNA structure and base-pairing rules.
- Explain why DNA replication is semiconservative.
- Compare transcription and translation.
- Use codons to connect mRNA sequence to amino acid sequence.
- Explain how mutation can alter protein and trait outcomes.

OBJECTIVE 1
Describe DNA structure and base-pairing rules.

OBJECTIVE 2
Explain why DNA replication is semiconservative.

OBJECTIVE 3
Compare transcription and translation.

OBJECTIVE 4
Use codons to connect mRNA sequence to amino acid sequence.

OBJECTIVE 5
Explain how mutation can alter protein and trait outcomes.

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Topic sequence

- DNA structure: DNA stores information in nucleotide sequences and complementary base pairing.
- Replication: Replication copies DNA using each strand as a template.
- Transcription: Transcription produces RNA from a DNA gene.
- Translation: Translation uses mRNA codons and ribosomes to build proteins.
- Mutation and gene expression evidence: Mutations and expression patterns connect molecular changes to traits.

01

DNA structure

DNA stores information in nucleotide sequences and complementary base pairing.

02

Replication

Replication copies DNA using each strand as a template.

03

Transcription

Transcription produces RNA from a DNA gene.

04

Translation

Translation uses mRNA codons and ribosomes to build proteins.

05

Mutation and gene expression evidence

Mutations and expression patterns connect molecular changes to traits.

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Module 07: Molecular Genetics Evidence Map

- Module focus: Molecular Genetics
- Central claim: Molecular Genetics explains DNA information flow from sequence to RNA, protein, and trait evidence.
- Key nodes to connect: DNA structure, Replication, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-07_molecular-genetics.md supplies an evidence surface for the map.

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Module 07: Molecular Genetics Reasoning Sequence

- Module focus: Molecular Genetics
- Trace the model: DNA structure -> Replication -> Transcription -> Translation
- Inputs become outputs: DNA structure, Replication -> Describe DNA structure and base-pairing rules., Explain why DNA replication is semiconservative.
- Feedback check: lab-07_molecular-genetics.md evidence checks whether students can explain transcription.
- Lab connection: lab-07_molecular-genetics.md gives students a process to observe or test.

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Terms as evidence handles

- DNA: Nucleic acid that stores hereditary information.
- Nucleotide: DNA or RNA subunit with sugar, phosphate, and base.
- Replication: Process that copies DNA.
- Transcription: Synthesis of RNA from a DNA template.
- Translation: Synthesis of protein from mRNA codons.
- Codon: Three-base mRNA sequence read during translation.

DNA
Nucleic acid that stores hereditary information.

NUCLEOTIDE
DNA or RNA subunit with sugar, phosphate, and base.

REPLICATION
Process that copies DNA.

TRANSCRIPTION
Synthesis of RNA from a DNA template.

TRANSLATION
Synthesis of protein from mRNA codons.

CODON
Three-base mRNA sequence read during translation.

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Lab connection

- Lab file: lab-07_molecular-genetics.md
- Lab evidence should test or illustrate: Transcription
- Students should leave with a concrete observation, comparison, or model tied to the module claim.

QUESTION
DNA structure

EVIDENCE
lab-07_molecular-genetics.md

CLAIM
Describe DNA structure and base-pairing rules.

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Contrast check

- Do not stop at: DNA stores information in nucleotide sequences and complementary base pairing.
- Stronger explanation: Mutations and expression patterns connect molecular changes to traits.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE
DNA stores information in nucleotide sequences and complementary base pairing.

DEEPER
Mutations and expression patterns connect molecular changes to traits.

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Module 07: Molecular Genetics Retrieval and Lab Check

- Module focus: Molecular Genetics
- Retrieval prompt: How does DNA structure store information?
- Answer check: Describe DNA structure and base-pairing rules.
- Required terms: DNA, Nucleotide, Replication
- Lab connection: lab-07_molecular-genetics.md; lab-07_molecular-genetics.md supplies evidence for sequence, transcription, translation, and gene-expression evidence.

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Practice quiz bridge

- 1. In DNA, the base adenine always pairs with:
- 2. DNA replication is described as "semiconservative" because each new double helix contains:
- 3. Which sequence shows the normal flow of genetic information?
- 4. A mutation changes a single DNA base. Why might this matter?

QUESTION 1

Answer first

In DNA, the base adenine always pairs with:

QUESTION 2

Answer first

DNA replication is described as "semiconservative" because each new double helix contains:

QUESTION 3

Answer first

Which sequence shows the normal flow of genetic information?

QUESTION 4

Answer first

A mutation changes a single DNA base. Why might this matter?

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Synthesis and exit ticket

- Exit claim: explain dna structure using evidence from lab-07_molecular-genetics.md.
- Must include: DNA, Nucleotide, Replication.
- Revision prompt: what evidence would change your explanation?

CLAIM
DNA structure

EVIDENCE
lab-07_molecular-genetics.md

REVISION
What would change your mind?